

Akhil Joshi

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Cloud Platform & DevOps Engineer | AWS/Azure, Kubernetes/EKS, Terraform, Kafka/MSK, GitHub Actions | Big-4 Alumni (PwC, Deloitte)

PROFESSIONAL SUMMARY

5+ years building enterprise-scale cloud infrastructure, designing and operating CI/CD systems in large, multi-account environments, spanning across Big 4 consulting through logistics platforms at Koch Industries. Track record of standardizing engineering workflows, reducing provisioning and deployment effort, lowering infrastructure cost, strengthening security, and improving production incident response. Awarded Koch Scholar. Active in cloud-native community through publications, presentations, judging, projects, and mentoring.

PROFESSIONAL EXPERIENCE

Cloud Platform Engineer | KBX Logistics (Koch Industries)

Jan 2025 - Present | Wichita, KS

- **Zero-Trust Kafka Architecture:** Own Amazon MSK/Kafka across 23 AWS accounts; designed IAM-authenticated, PrivateLink-only connectivity and topic-level RBAC for 12+ microservices, and operate Kafka Connect with containerized plugins and Datadog dashboards, monitors, and alerts for end-to-end connector health.
- **CI/CD Platform:** Led migration of application and infrastructure delivery from Azure DevOps to GitHub Actions and built reusable AWS OIDC workflows with Orca/AWS security gates, approvals and automated rollback to reduce manual recovery effort.
- **Terraform & Self-Service:** Standardized 15+ AWS environments using reusable Terraform modules, remote state, drift detection, gated promotion workflows, and GitHub Issue-based provisioning, saving ~2 engineering days per rollout.
- **Kubernetes Platform:** Operate multi-cluster EKS across 4 environments using Argo CD, Helm, KEDA, Karpenter, RBAC, TargetGroupBinding, and ingress patterns to standardize workload delivery, autoscaling, access, and service connectivity.
- **Production Reliability:** Own SLOs and recurring week-long on-call rotations for 6 distributed services, leading P1 response, RCA, postmortems, and corrective actions; Datadog improvements reduced MTI from 47 to 28 minutes.
- **Internal Developer Platform:** Built Python/YAML self-service tooling integrating AWS APIs, Terraform state, IAM guardrails, and runbooks, replacing 5+ recurring CLI workflows for 10+ engineers.

Cloud Platform Engineer Intern | KBX Logistics (Koch Industries)

May 2024 - Aug 2024 | Wichita, KS

- **EKS FinOps:** Replaced Cluster Autoscaler with Karpenter Spot NodePools across 12 EKS clusters, reducing node provisioning from ~4 minutes to ~90 seconds and lowering monthly EC2 compute spend by ~20%.
- **Application Performance:** Added ElastiCache Redis caching across 4 microservices, reducing average API latency by ~15% and deferring a planned RDS vertical-scaling event.
- **Platform Security:** Deployed OWASP Core Rule Set controls through Gloo Enterprise for internal platform endpoints, closing SQL-injection and path-traversal findings from a prior security review.
- **Kubernetes Observability:** Built Prometheus/Grafana dashboards for EKS node, workload, and ingress health, replacing ad hoc kubectl and log checks with standardized monitoring for deployment, scaling, and incident triage.

Graduate Assistant - Computer Science | Kansas State University

Jan 2023 - Dec 2024 | Manhattan, KS

- **Automation & Systems:** Built a Python/OCR pipeline that converted ACM CS Curricula 2023 content into structured JSON, reducing manual curriculum-analysis effort by ~40%; supported software-engineering and systems instruction for 45+ students per semester.

DevOps Engineer II | PricewaterhouseCoopers (PwC)

Jun 2021 - Nov 2022 | Bengaluru, India

- **AKS Delivery Blueprint:** Designed a reusable onboarding model for containerized applications using AKS, ACR, Key Vault, Managed Identity, Azure Monitor, and Application Insights, reducing manual environment and access configuration by ~40%.
- **Provisioning Acceleration:** Automated application-environment provisioning and dependency setup, cutting request-to-ready turnaround from 1–2 days to under 4 hours and removing repeated handoffs between development and operations teams.
- **Release Factory:** Standardized multi-stage Azure DevOps YAML pipelines across 4 services with artifact promotion, environment approvals, smoke validation, and rollback, reducing release-preparation effort by ~35%.
- **Pre-Production Defect Prevention:** Introduced static analysis, image scanning, secret validation, and deployment quality gates into delivery workflows, catching 15–20 security and configuration issues before production release windows.
- **Production Stabilization:** Led troubleshooting and corrective engineering for AKS deployment failures, pod crashes, API errors, and Azure dependency issues, reducing repeat release-blocking defects by ~25%.

- **AWS Infrastructure as Code:** Built reusable CloudFormation stacks for EC2, ALB, RDS, security groups, and environment configuration, standardizing infrastructure provisioning and reducing manual configuration drift across application environments.
- **Blue/Green Release Engineering:** Built CodeDeploy blue/green workflows that eliminated 45+ minutes of release downtime across 6 consecutive production deployments.
- **Linux & Deployment Automation:** Built Bash and CodeDeploy lifecycle checks for pre/post-deployment validation, service health verification, and rollback triggers, reducing manual release verification and preventing unhealthy traffic cutovers.
- **Observability & Operations:** Built CloudWatch dashboards and composite alarms for ALB 5xx, target health, API p95 latency, RDS CPU, and connection saturation, replacing manual release-window checks with proactive monitoring.
- **Database Reliability:** Optimized PL/SQL procedures and indexing across 6 high-traffic tables, cutting peak database latency by ~40% and eliminating 2 recurring SLA-impacting timeout incidents per quarter.

TECHNICAL SKILLS

- **Cloud & Platform Engineering:** AWS (EKS, MSK/Kafka, EC2, S3, IAM, VPC, PrivateLink, Route 53, ALB/NLB, RDS, ElastiCache, CloudWatch, KMS, Secrets Manager, Lambda, EventBridge, SNS, CodeDeploy); Azure (AKS, Azure DevOps, Key Vault, Managed Identity, ACR, Azure Monitor, Application Insights, Azure SQL, Blob Storage, Entra ID); platform engineering, developer self-service, FinOps
- **Kubernetes & Containers:** Kubernetes, EKS, AKS, Docker, Helm, Argo CD, GitOps, Karpenter, KEDA, Auto Scaling Groups, TargetGroupBinding, RBAC, ingress, Gloo Enterprise, workload autoscaling, multi-cluster operations
- **Infrastructure as Code & Automation:** Terraform, reusable modules, remote state, state management, drift detection, CloudFormation, GitHub Issues self-service, Python/YAML automation, infrastructure provisioning, configuration standardization
- **CI/CD & Release Engineering:** GitHub Actions, Azure DevOps, Harness, reusable workflows, pipeline-as-code, AWS OIDC, artifact promotion, approval gates, canary and blue/green deployments, smoke tests, automated rollback, release governance
- **Reliability & Observability:** Datadog APM/Logs/Alerting, Prometheus, Grafana, CloudWatch, Azure Monitor, SLI/SLO, on-call, P1 incident response, RCA, postmortems, runbooks, MTTI/MTTR, high availability, autoscaling, reliability remediation
- **Security, Identity & Networking:** IAM, cross-account IAM, RBAC, least privilege, AWS OIDC, KMS, Akeyless Vault, Secrets Manager, WAF, OWASP CRS, Zero Trust, SAST, container-image scanning, VPC routing, PrivateLink, security groups, Route 53/DNS, TCP/IP, HTTP/TLS, ALB/NLB
- **Programming, Systems & Data:** Python, Bash/Shell, Java, SQL, YAML, JSON, Linux, Git/GitHub, ServiceNow, Salesforce, Kafka, event-driven architecture, microservices, REST, gRPC, Redis, RDS, PostgreSQL, MySQL, MongoDB, PL/SQL, Apache Airflow

RESEARCH & PUBLICATIONS - IEEE & MDPI

8+ peer-reviewed papers in cloud security, Kubernetes runtime defense using eBPF/ML and LLM-based IaC automation.

- SafeFix-Agent: Diff-Only LLM Remediation for Infrastructure-as-Code with Policy, Test, and Sandbox Gates
- KubeShield: Multi-Modal Anomaly Detection for Kubernetes Using Ensemble Learning
- Measuring Cloud Misconfigurations in the Wild: An Empirical Multi-Cloud Study

Additional publications on [Google Scholar](#)

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (SAA-C03)
- Microsoft Certified: Azure Developer Associate (AZ-204)
- Microsoft Certified: Information Protection Administrator (SC-400)
- GitOps Certified Practitioner (GitOps Enterprise)
- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Fundamentals (AZ-900)

EDUCATION

Master of Science, Computer and Information Science | Kansas State University, Manhattan, KS | 2023 - 2024

Bachelor of Technology, Computer Science & Engineering | Jawaharlal Nehru Technological University, Hyderabad, India | 2016 - 2020

HONORS & COMMUNITY

- Koch Scholar (2023) - competitive merit award for high-potential engineers within Koch Industries
- State of CD Research Contributor 2026 - Octopus Deploy
- Cloud-native community: KubeCon, HashiConf, IaCConf, AWS re:Invent; conference reviewer and program-committee contributor
- College Ambassador, JNTU Hyderabad (2019-20) | Student Contributor, Computer Society of India (2018-20)